Master’s Thesis : A Retrofit Sit-to-Stand Assistive System

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IIT Kanpur (B.Tech) — IISc Bangalore (M.Des, Product Design, Development & Management)

Introduction

Hello, I am ***Mudit Chand Narayan***, I was trained at ***IIT Kanpur in civil engineering,*** then ***at IISc Bangalore in Product Design and Manufacturing.***

I'm drawn to design driven by ***analysis*** — I like to gain a large share of my design experience through field work rather than the studio alone. My forte is ***mechanical validation*** and ***field-based requirement gathering*** including taking a concept, testing it against real constraints, and being willing to kill it or rebuild it when the numbers disapprove of the possible results.

What drew me towards this role was concept-to-spec product design grounded in engineering constraints.

1

IIT Kanpur

B.Tech, Civil Engineering

Core engineering fundamentals: mechanics, structural analysis, materials, quantitative problem-solving under constraints.

2

IISc Bangalore

M.Des, Product Design, Development & Management

Housed within the Dept. of Design & Manufacturing as an extension of Mechanical Engineering. Coursework and thesis work were engineering-heavy, including kinematics, torque analysis, DFM, prototyping.

3

Net result

A designer who validates with numbers

Concepts aren't chosen on aesthetics alone — they're screened through static analysis, mechanical checks, and physical falsification before being carried forward.

Signature Project

A Retrofit Sit-to-Stand Assistive System

M.Des Thesis — with Joe Thomas — IISc Bangalore, Dept. of Design & Manufacturing

Field Research & Need Analysis → **Concept Generation** → **Engineering Validation** → **Embodiment Design** → **Physical Prototyping** → **Manufacturing Specification**

Developed four concepts, mathematically evaluated each against torque/force requirements, killed three on engineering grounds, then took the survivor through embodiment to physical prototype.

600 Nm

Assistive torque estimated & validated

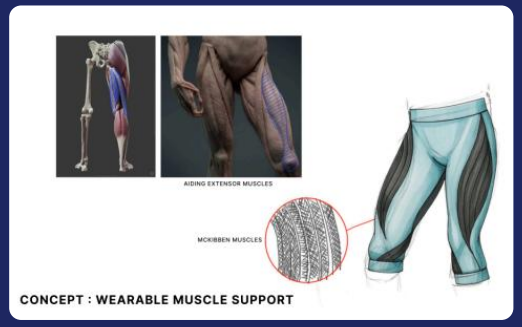
45000 ₹

Target market price achieved

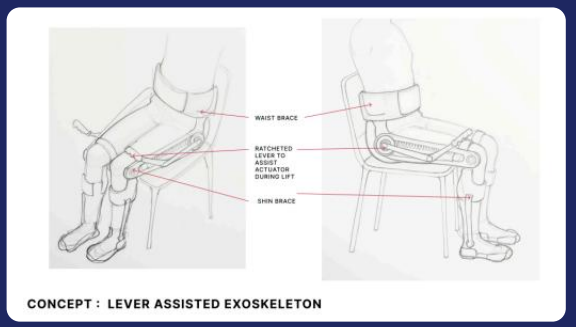
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Prototypes built — each falsified something specific







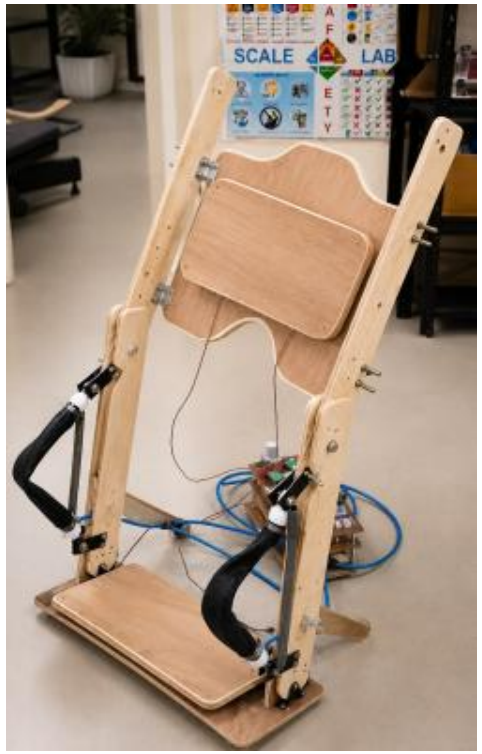


Signature Project

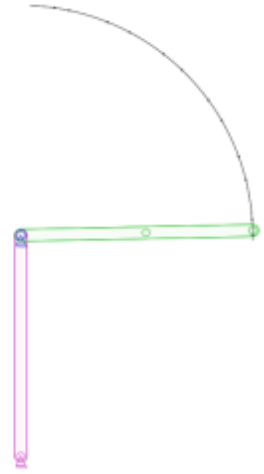
A Retrofit Sit-to-Stand Assistive System

Four concepts were generated for elderly sit-to-stand assistance — each evaluated against calculated torque and force requirements before selection.

Wearable inflatable / pneumatic muscle support	Rejected	Torque math showed ~600 Nm needed; pneumatic muscles at 40 psi delivered ~78 N would require impractical bulk or pressure.
Localized knee-assist mechanism	Rejected	Concentrated loading at the knee introduced instability and misalignment with natural joint trajectory.
Wearable fluidic soft exosuit	Rejected	Shear forces of 70–100 N at the skin interface — unacceptable for elderly users with reduced tissue resilience.
Externally stabilized four-bar linkage + cable actuation	Selected	Distributed loads across an external frame; matched natural CoM trajectory; lowest shear, highest stability.



Prototype 1



CoM trajectory

Experiences :

- Manufacturing Processes, i.e. ***CNC/CHS tube fabrication, welding, powder coating, injection moulding, stamping***.
- Using ***Innovation Solution Questionnaire (ISQ)*** and Habitat Study to frame user needs.
- Hands-On experience with analytical design methods like ***FMEA, QFD, TRIZ***.
- 4 Years of experience with ***CAD software*** such as Fusion360 and FreeCAD.
- Basic Knowledge of Operations Management processes like CPM and PERT.
- Topology Optimisation.

Diagnosing and Fixing a Mechanical Failure

Prototype 1

FAILED

First-class lever mechanism.

Failure: 1:4 mechanical disadvantage against the actuator, plus a single circular-arc coupler path that lost contact with the user mid-transfer.

Prototype 2

PARTIALLY FAILED

Planar four-bar linkage — kinematics validated, but transmission failed.

Diagnosis: Sliding-contact cable routing introduced significant friction losses; the Capstan relationship (suggested by our mentor) showed that the required input tension increased exponentially with $\mu\theta$.

Final Design

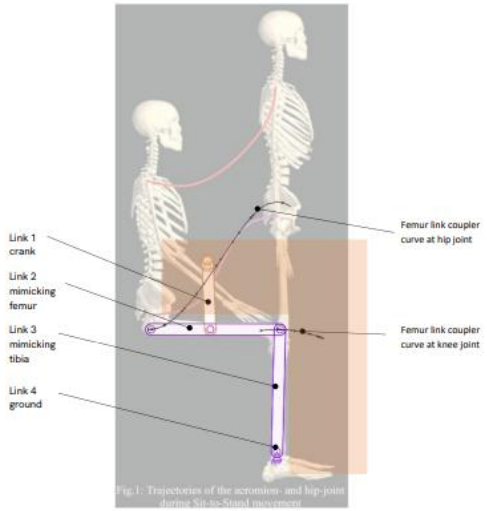
CORRECTED

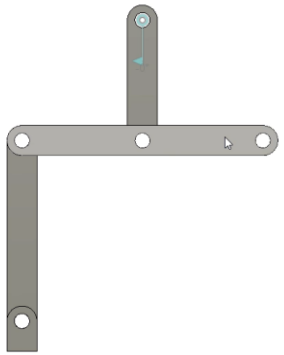
Same four-bar geometry, rolling-bearing sheave pulleys replacing sliding contacts.

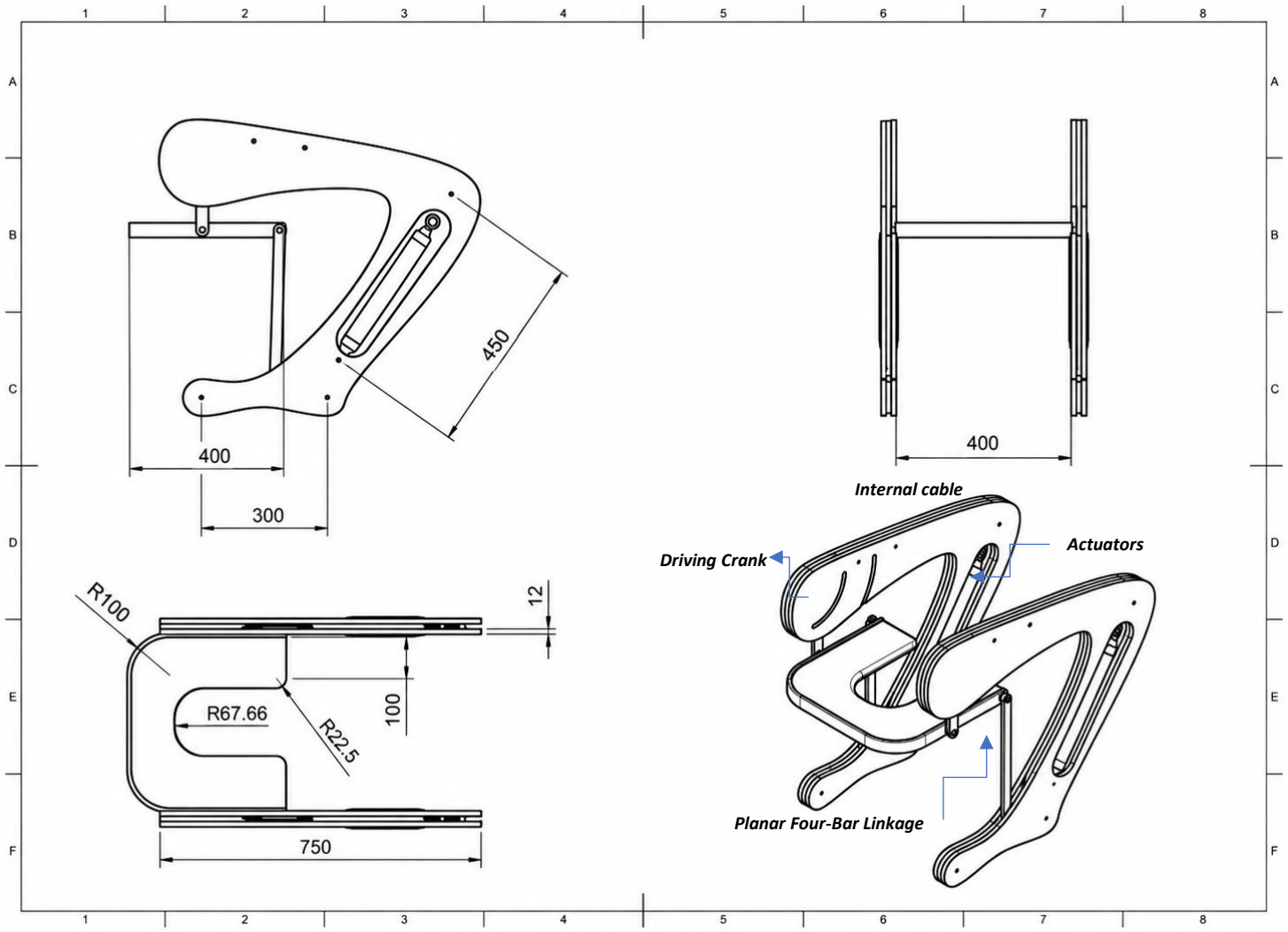
Result: friction reduced by enough to bring the tension safely within the actuator's operating envelope.

Signature Project

A Retrofit Sit-to-Stand Assistive System : Detailed Design







Measurements are in mm

Material Choices



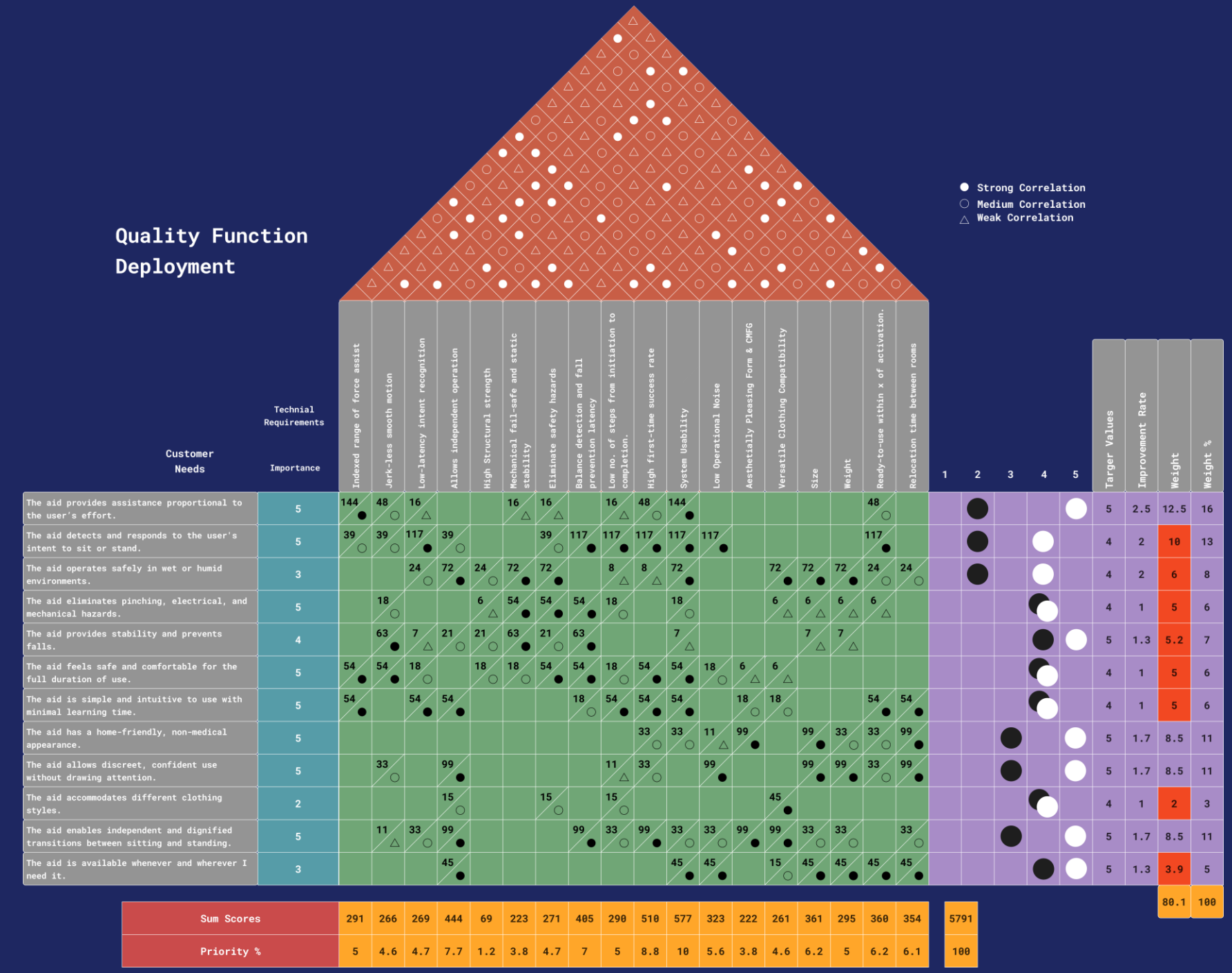
"Biophilia hypothesis," suggests humans experience therapeutic effects from natural materials.

Temporal continuity:
materials that existed in their formative years serve as indexical signs connecting present to past.

Signature Project

A Retrofit Sit-to-Stand Assistive System : QFD

Quality Function Deployment



Customer Needs	Technial Requirements: Importance	Indexed range of force assist	Jerk-less smooth motion	Low-latency intent recognition	Allows independent operation	High Structural strength	Mechanical fail-safe and static stability	Eliminate safety hazards	Balance detection and fail prevention latency	Low no. of steps from initiation to completion.	High first-time success rate	System Usability	Low Operational Noise	Aesthetially Pleasing Form & CMFG	Versatile Clothing Compatibility	Size	Weight	Ready-to-use within x of activation.	Relocation time between rooms	1	2	3	4	5	Targer Values	Improvement Rate	Weight	Weight %
The aid provides assistance proportional to the user's effort.	5	144 ●	48 ○	16 △			16 △	16 △		16 △	48 ○	144 ●						48 ○			●			○	5	2.5	12.5	16
The aid detects and responds to the user's intent to sit or stand.	5	39 ○	39 ○	117 ●	39 ○			39 ○	117 ●	117 ●	117 ●	117 ●	117 ●					117 ●			●		○		4	2	10	13
The aid operates safely in wet or humid environments.	3			24 ○	72 ●	24 ○	72 ●	72 ●		8 △	8 △	72 ●			72 ●	72 ●	72 ●	24 ○	24 ○		●		○		4	2	6	8
The aid eliminates pinching, electrical, and mechanical hazards.	5		18 ○			6 △	54 ●	54 ●	54 ●	18 ○		18 ○			6 △	6 △	6 △	6 △					◐		4	1	5	6
The aid provides stability and prevents falls.	4		63 ●	7 △	21 ○	21 ○	63 ●	21 ○	63 ●			7 △				7 △	7 △						●	○	5	1.3	5.2	7
The aid feels safe and comfortable for the full duration of use.	5	54 ●	54 ●	18 ○		18 ○	18 ○	54 ●	54 ●	18 ○	54 ●	54 ●	18 ○	6 △	6 △								◐		4	1	5	6
The aid is simple and intuitive to use with minimal learning time.	5	54 ●		54 ●	54 ●				18 ○	54 ●	54 ●	54 ●		18 ○	18 ○			54 ●	54 ●				◐		4	1	5	6
The aid has a home-friendly, non-medical appearance.	5										33 ○	33 ○	11 △	99 ●		99 ●	33 ○	33 ○	99 ●			●		○	5	1.7	8.5	11
The aid allows discreet, confident use without drawing attention.	5		33 ○		99 ●					11 △	33 ○		99 ●			99 ●	99 ●	33 ○	99 ●			●		○	5	1.7	8.5	11
The aid accommodates different clothing styles.	2				15 ○			15 ○		15 ○					45 ●								◐		4	1	2	3
The aid enables independent and dignified transitions between sitting and standing.	5		11 △	33 ○	99 ●				99 ●	33 ○	99 ●	33 ○	33 ○	99 ●	99 ●	33 ○	33 ○		33 ○			●		○	5	1.7	8.5	11
The aid is available whenever and wherever I need it.	3				45 ●							45 ●	45 ●		15 ○	45 ●	45 ●	45 ●	45 ●				●	○	5	1.3	3.9	5
																											80.1	100
Sum Scores		291	266	269	444	69	223	271	405	290	510	577	323	222	261	361	295	360	354	5791								
Priority %		5	4.6	4.7	7.7	1.2	3.8	4.7	7	5	8.8	10	5.6	3.8	4.6	6.2	5	6.2	6.1	100								

Relevant Engineering Experiences

Conceptual & detailed design (mandatory)	Thesis: 4 concepts screened via engineering analysis; final embodiment detailed to component level
New Product Development	Full field-research → concept → embodiment → prototype → manufacturing-spec cycle
Assembly design & mechanical fundamentals	Four-bar bell-crank linkage synthesis; diagnosed and corrected a transmission failure
Manufacturing process knowledge	Factory-floor process mapping: welding, powder coating, moulding, stamping
3D CAD (SolidWorks preferred)	4 years of working with Fusion 360 & FreeCAD, experienced with parametric modelling, assemblies and mechanical design.
Project & cross-functional collaboration	Two-person thesis team with faculty and hospital stakeholders; factory-floor interviews for the chair project

Thank You

Supplementary Data (Not for presentation) :

29

CONCEPT 6

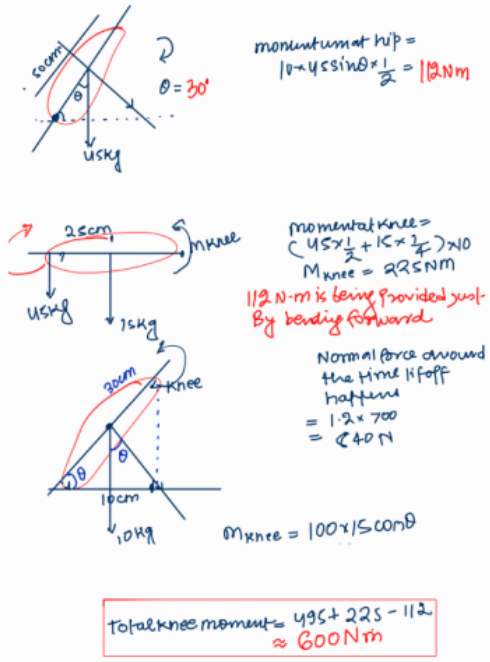
Scan	Validate	Segregate	Collect	Authenticate
Make it a QR code for scanning.	Have an address database with all possible addresses.	Colour them according to the destination.	Robotic arm picks it up and puts it somewhere.	2 step verification
Camera based identification setup.	Bio-metric validation.	Scanning the item based on pincodes.	Self-Validation	Non replicable
Scanning glasses.	Digitally printed label, can be sorted on the route-map.	AI-based OCR	Centralized system	Tampering assessment
Self-aligning scanner head	Add geographical plus code	Have different stamps, based on destination.	Standardization	Compact/ fitting

Example of a Morphology Matrix used during the conceptualization of one of the solutions for the Post Office Distribution System

Supplementary Data (Not for presentation) :

THE SIT STAND PROBLEM

Insights from Mathematical Simulations



Assuming we are to provide 600NM for 60 cycles a day

$N_{cycle} = 60$

$\eta = 80\%$

angular displacement $= \pi/2$ rad

angular speed $= \omega = \frac{2\pi}{3} \approx 20$ RPM

Mechanical energy Required $= T\theta$

$= 600 \times \frac{\pi}{2}$

$= 942$ Joules

Total energy for 60 cycles $= 56 \times 10^3$ Joules

Mechanical power $= \frac{56 \times 10^3}{180}$

$= 300$ watt

Design Power Required $= \frac{300}{0.8} = 375$ watt

Current $i = \frac{P}{V} = 15.625$ Amp

Basic Assumptions →

Weight = 70kg
Height = 180 cm
Torso height = 100cm
Thigh length = 50cm
Shin length = 30cm

torso weight = 45kg
thigh weight = 15kg
shin weight = 10kg

CONCLUSIONS FROM STATIC ANALYSIS-

- The net torque required to raise falls around ***600N-M.***
- ***Bending the trunk forward*** supplies a major portion of required torque.
- We would require a motor that can provide ***large torque at small rotation speeds***.
- Required battery power is around ***5000mAH.***

Calculations for the required knee moment

Supplementary Data (Not for presentation) :

Insights from Mathematical Simulations

Force generated by McKibben Muscles: –

$$F = \frac{\pi D_0^2 P}{4}(3\cos^2\theta - 1)$$

$D = 12mm$
$P = 40psi$
$\theta = 30°$

$40psi = 275.79kPa$

$$F = \frac{\pi(0.012)^2 \times 275790\ (1.25)}{4}$$

$F = 38.9$ Newtons

Forces generated on the thigh clamp:

Radial force $= 2 \times F \times \sin 45$
$= 54$ Newtons

Shear force at the skin = $F_{contraction}$
$f = \mu F_{contraction}$ $\mu = 0.5$
$F_c = 38 \times 2 = 76$ Newton

Shortening of length in a mckibben Muscle

$$L = L_0\sqrt{\frac{b^2 - \cos^2\theta}{b^2 - \cos^2\theta_0}}$$

$$b = \frac{\pi D_0 n}{L_0}$$

$L_0 = 50cm$
$D_0 = 12mm$
$n = 1000$

$b = 75.39$

$L = 44cm$

$\Delta L = 6cm$
Shortening



CONTRACTION

- ***A consistent contraction of 5cm*** was observed during empirical tests, which aligns very well with the theoretical shortening.
- A design with ***wider muscle diameter would produce more force*** with the cost of discomfort.
- The ***shear forces*** against the skin range around 70-100N.
- All the force generated by contraction ***directly goes to the knee.***

Mathematical Simulation of One of the concepts and Provided Technical reason for the rejection

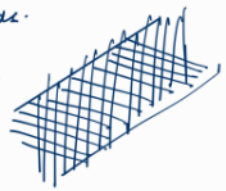
Supplementary Data (Not for presentation) :

Insights from Mathematical Simulations

In case of Using Mckibben Muscles:-

Force is applied on Both ends.

θ = Braid angle w.r.t. the muscle Axis
b = Length of one Braid strand
n = Number of turns along length
D = Instantaneous Diameter
L = Instantaneous length



geometric Relationship = $b^2 = L^2 + (\pi D n)^2$

Force generation Equation = $F = \frac{\pi D^2 P}{4}(3\cos^2\theta - 1)$

$F \propto D^2$

$F \propto P$

Force Reversal happens at $\theta = 54.7°$

For D = 5mm
L = 60cm
θ = 30°
P = 0.5mpa

$F = 12.2718\ N$

P is Reported to be at max = 0.7

Studies from a paper

Theoretical Contraction

$$F = \frac{\pi}{4} D^2 P \left(\frac{1}{\sin\theta}\right)^2 \left\{3(1-\varepsilon)^2 \cos^2\theta - 1\right\}$$

ε = ratio diameter of the tube

Changing the Fibers with polyester & PBO
PBO provides tensile strength 5.8 GPa
Generating the force of 25KN

Sleeve structure			Bundles	diex
C –	PBO Fiber	25°	95	6600
D –	PBO Fiber	30°	95	6600

FORCE GENERATION

- The force produced by a pneumatic muscle is ***proportional to the pressure applied.***
- The force produced by a pneumatic muscle is ***proportional to the square of diameter of muscle.***
- A pneumatic muscle can sustain around ***150PSI*** of pressure.
- The force generated by a muscle largely ***varies on diameter and material.***



Each singular embedded muscle provided 10N Net force